

Riley Beenders

SOFTWARE-INTEGRATED SYSTEMS | SELF-HOSTED PLATFORMS | ENGINEERING LEADERSHIP

Bethlehem, PA | [\(484\) 523-4431](tel:(484)523-4431) | rileybeenders@gmail.com | www.rileybeenders.com | linkedin.com/in/riley-beenders

PROFESSIONAL SUMMARY

Engineering and technology leader with 10+ years of multidisciplinary experience and progressive responsibility through CTO and Lead R&D roles. Designs, integrates, tests, and supports software-connected industrial systems spanning self-hosted infrastructure, containerized applications, remote operations, high-frequency data acquisition, controls, and hardware. Leads cross-functional teams and external partners from requirements and architecture through validation, commercialization, and continuous improvement.

CORE EXPERTISE

Software & Data: Python; LabVIEW; TypeScript/React/Next.js (Node.js runtime); Git/GitHub; Modbus; CAN bus; high-frequency data acquisition; analytics; application testing and debugging

Infrastructure & Architecture: Docker; Kubernetes; self-hosted infrastructure; Proxmox virtualization; Windows/Linux administration; server networks and VLANs; reverse-proxy and firewall routing; platform standardization

Leadership, AI & Delivery: Requirements definition; software/hardware integration; AI-assisted monitoring integration; LM Studio; oMLX; technical strategy; risk analysis; technical documentation; cross-functional/global leadership; external software-partner collaboration

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Lead the East Coast engineering team and collaborate across Proteor's global portfolio, owning software-integrated products from requirements and architecture through integration, validation, commercialization, risk mitigation, and support.
- Lead ProteorPrint OS implementation with an external software partner, translating machine requirements into a network-accessible remote operations and diagnostics interface with live video, AI-assisted alerts, and production metrics.
- Developed a LabVIEW/Modbus metrology application that collects nine data points every 15 ms from laser sensors, displays live tolerance charts, generates PDF reports, and stores traceable records on a remote server.
- Oversee Microsoft Business Central implementation for division products and items, synchronizing sales and inventory data across global Proteor locations while standardizing internal software and processes.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Progressed through four roles to CTO, directing product development, internal IT infrastructure, and technology strategy while remaining hands-on across firmware, software, electrical, and manufacturing systems.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM that centralized orders, inventory, tasks, projects, and customer interactions and automated dynamic BOM generation linked to purchasing.
- Integrated a modular multi-board CAN-bus architecture and independently testable electronics enclosures, simplifying wiring, testing, and debugging while reducing installation to minutes.
- Led full-lifecycle delivery of large automated systems across architecture, assembly, firmware tuning, calibration, operator training, and post-delivery technical support.

ADDITIONAL RELEVANT EXPERIENCE

RileyBeenders.com - Full-Stack Portfolio Project | 2026 | Built and deployed a TypeScript/React/Next.js interactive resume with structured JSON content, a Node.js PDF endpoint, Git version control, and web analytics.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Built network infrastructure, deployed 750+ wireless access points, and automated setup-data entry for 2,500+ Chromebooks with an Arduino tool.

Blue Mountain Ski Resort - Ski Lift Operator | 2019 - 2021 | Performed motor/control safety checks and supported maintenance on guest-facing passenger lifts while assisting thousands of guests.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: MathWorks Machine Learning, MATLAB, Simulink, and Simscape Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)